Algorithm	Upper Bounding (UB)					Branching					Reduction						
	S-Based	R-Based	SR-Based	UBR1	UBR2	Binary	Pivot	S-Based	R-Based	SR-Based	Two-Hop	RR1	RR2	RR3	CTCP	BR1	BR2
U-MkP (Our Work)	✓	✓	✓		✓		✓	✓	✓	✓	✓ [g]	✓	✓	✓	✓ [T]	✓	✓
kPlexT (SIGMOD'24)					✓	✓					✓ [g]	✓	✓	✓	✓ [T]	✓	✓
Maple (IJCAI'23)				✓			✓				✓ [g]	✓	✓	✓	✓ [+]	✓	✓
Seesaw (arXiv'23)	✓ [+]	✓ [+]	✓	—	—	—	—	—	—	—	—	—	—	—	—	—	—
DiseMKP (IJCAI'23)	✓ [+]							✓ [+]				✓	✓	✓	✓ [T]		
kPlexS (PVLDB'22)				✓		✓					✓ [g]	✓	✓	✓	✓ [+]	✓	✓
KpLeX (IJCAI'21)	✓							✓				✓	✓	✓			
Maplex (AAAI'21)		✓				✓					✓	✓	✓	✓	✓ [T]		
BnB (IJCAI'18)						✓						✓	✓	✓			
BS (AAAI'17)							✓				✓	✓	✓	✓			✓

*** Note:** ✓ [g]: Two-hop graph constructed; ✓ [+]: Improved version; ✓ [T]: Top-level only

— : Seesaw only proposes UB rules, no open-source code;